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Full Length Article

Supplemental feeding on rangelands: New dynamics of the livestock in the El Ouara rangelands in southern Tunisia

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ABSTRACT

Fodder production in arid rangelands has been traditionally considered as the main source of nutrition for livestock. However, the production of these fodder may be influenced by some socio-economic and climatic factors. This study aims to identify the various rangeland use forms and determine the types of livestock feeding system in the El Ouara rangelands of Ben guarden District in southern Tunisia by analysing the importance of supplemental feeding and main determinants of this practice. Data were collected by a survey that targeted a sample of 50 breeders in the El Ouara rangelands. Descriptive statistics and binary logistic regression were used to analyse the data. The results showed that there are two types of association practiced for breeders:"Cherka" association practiced by 10.00% of breeders and cash payment association practiced by 53.00% of breeders. Then, transhumance, sedentary, and semi-urban forms are the main ways in which breeders use the rangeland. There are four livestock feeding systems in the El Ouara rangelands being used to feed livestock: natural rangeland, rangeland-based, mixed, and concentrate-based feeding systems. Supplemental feeding has become fundamental for the livestock feeding system in the El Ouara rangelands. The breeders with the largest camel size still use the natural rangeland feeding system, while the breeders who have the largest herd size practice the mixed feeding system. The logistic regression results reveal that breeder age, cereal area, herd size, migrant remittance, second activity, and agricultural development group membership can significantly impact the supplemental feeding. In summary, this study can help policy-makers plan innovative practices based on climatic change, ensure the sustainability of livestock feeding system, and make effective decisions for local development.

1. Introduction

Herding people use natural rangelands as their life support systems in which they keep diverse livestock, usually sheep, goats, and camels. These rangelands account for around 80.00%–85.00% of their feed needs. In African countries, the contribution of rangelands to domestic livestock feed needs is over 85.00% (Zaied et al., 2020). In Tunisia, large arid rangelands constitute more than two thirds of the total rangeland area, i.e., about 5.5×10^6 hm^2, of which livestock is the main activity on pastures (Ben Salem, 2011). Ruminants have contributed to the development of pastoral resources and socio-economy in arid regions (Gaddour et al., 2014).

However, climate change has led to a decrease in fodder resources and the depletion of water sources, thereby affecting the productivity of natural rangelands. Indeed, any decrease in precipitation will reduce the productivity of natural rangelands (Djohy et al.,

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2022). The livestock feeding system and the fodder resources available on rangelands are largely interdependent. Thus, in the past, the numbers of ruminants fluctuated considerably depending on the fodder resources of rangelands, which lie on precipitation. Since the introduction of supplemental feeding and its promotion to all types of livestock, the fluctuation in livestock numbers have been less remarkable, especially with the implementation of the program to safeguard livestock during drought periods and subsidy granted by the state. Therefore, since the 1990s, the number of small ruminants has been increasing regularly, which is different from the past due to new feeding habits (Abaab et al., 2021).

The scarcity and instability of these fodder resources, coupled with the endogenous factors of breeders (manpower, financial resources, socio-cultural characteristics, etc.) have led to a change in feeding systems. In order to cover the needs (in milk and meat) of the population, new livestock feeding system has emerged, namely peri-urban livestock feeding system (Ayoub, 2021). Supplemental feeding is a practice that appeared after a shortage of fodder on rangeland (Nasr et al., 2000; Bourbouze, 2006; Ben Saad and Bourbouze, 2010; Rjili et al., 2023). In arid areas of Tunisia, new livestock farming management types have been detected on rangelands (Najari et al., 2011; Bourbouze, 2018). Nevertheless, maintaining the systems relies on a thorough understanding of the different forms of exploiting rangelands to identify their weaknesses and to safeguard the lands of common rangelands from degradation. Consequently, the proposes of this study are to (1) characterise the existing livestock feeding system on common rangelands by identifying the different livestock guarding methods and the main rangeland use forms; (2) identify and analyse the livestock farming feeding system applied by breeders to ensure the sustainability of their farming system when facing scarce feed resources; and (3) determine the factors that affect the supplemental feeding on rangelands.

2. Materials and methods

2.1. Study area

The study was conducted in the El Ouara rangelands (32°12'00"N–32°47'52"N, 11°09'58"E–11°13'35"E; Fig. 1) of Ben guarden District in the Medenine Governorate (southern Tunisia) on the border with Libya. The region is characterised by aridity and articularly windy, which is subject to hot climate influences. The temperature is mild in winter (average temperature of 8.4°C in January) and very high in summer (average temperature of 37.2°C in August). The average annual precipitation is 131 mm (Climate-data, 2023). These hot winds, particularly "sirocco", have led to erosion and an increase in potential evaptranspiration (ETP). The particularity of this region

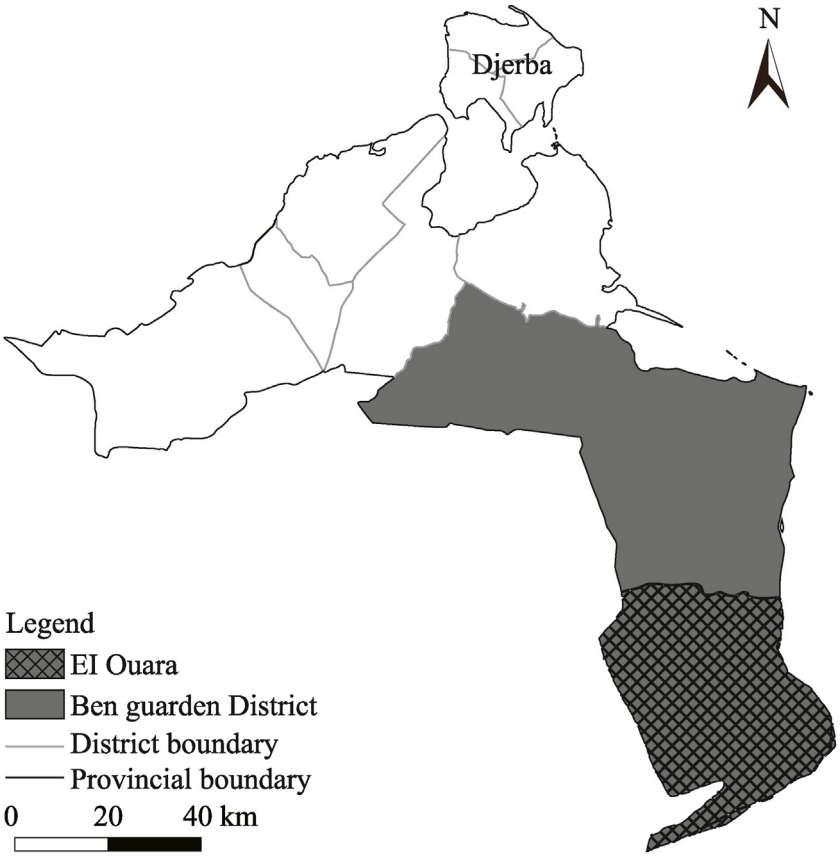

Fig. 1. Location of the El Ouara rangelands of Ben guarden District in the Medenine Governorate in southern Tunisia.

lies in the practice of extensive livestock feeding system on natural rangelands, which allows available space to be developed and the arid environment to be exploited. As the acceleration of privatization of common land, rainfed olive tree cultivation has increasingly developed to the detriment of rangelands (Ben Salem, 2011).

2.2. Data collection

Data were acquired by combining two techniques including documentary research and surveys, which were recommended by several researchers (Capillon, 1985; Dobremez and Bousset, 1996). The sample was based on a list that included 110 breeders who use the El Ouara rangelands of. Personnel contact (phone number) was made with 72.00% of them under the help of local stakeholders from the Ministry of Agriculture (the local extension cell and the livestock rangeland office of Ben guarden District). Face-to-face interviews were held with 50 breeders, and the survey was conducted between January and February in 2019 to collect information about breeders and their livestock farming socioeconomics. To meet our survey objectives, our problems had to be reformulated as understandable questions. Thus, questions were used as objectively as possible to avoid influencing respondents' answers. To ensure consistent progress, we divided the survey into five sections: (1) socio-demographic characterisation of the farm household; (2) the features of livestock farming structures and practices; (3) livestock feeding system and rangeland management method; (4) institutional and organisational aspects; and (5) the use of and access to rangelands.

The collected data were subjected to manual and computer analyses. All the statistical analyses were performed with the XLSTAT statistical software (Addinsoft, Paris, France).

2.3. Data analysis

We conducted the study in three steps following a progressive approach.

In the first step: descriptive statistics were used to explore the characteristics of livestock feeding system in the El Ouara rangelands and the socio-demographic characteristics of households, labour, herd management, and different rangeland use forms. Statistical analyses were performed with the XLSTAT statistical software.

In the second step: Principal Components Analysis (PCA) and Hierarchical Cluster Analysis (HCA) of Ward were used to identify and analyse the different livestock feeding systems followed by breeders on rangelands. We selected the variables to characterise livestock feeding system based on the seasonal changes in flock feeding management and the cost of each ingredient. Twenty-one feeding variables were introduced into the PCA, which were categorised into categorical and continuous groups (Table 1).

The principal components of the livestock feeding system were determined and the HCA of the first five principal components was conducted. Ward's HCA is one of the most widely used clustering techniques (Ferreira and Hitchcock, 2009). The final number of clusters was based on the loss of inertia in each partitioning of clusters. The interpretability of the results confirmed the number of clusters acquired by the HCA.

Finally, the different factors influencing the supplemental feeding practices were identified. Hence, breeders' decision to practice herd supplementation or not was considered to be a dichotomous variable. Thus, binomial logistic regression was applied (McFadden, 1973). The decision to livestock supplement feeding was used as a dependent variable (Y). The general logistic regression model was as follows:

Table 1
Descriptive statistics of the selected variables.

Description of variable	Type of variable	Percentage of breeders who indicated yes (%)
Use of barley in autumn in the total flock diet (1 = yes, 0 = no)	Categorical	75.61
Use of bran in autumn in the total flock diet (1 = yes, 0 = no)		43.90
Use of straw in autumn in the total flock diet (1 = yes, 0 = no)		31.71
Use of natural rangeland in autumn (1 = yes, 0 = no)		46.34
Use of barley in winter in the total flock diet (1 = yes, 0 = no)		12.20
Use of bran in winter in the total flock diet (1 = yes, 0 = no)		7.32
Use of natural rangeland in winter (1 = yes, 0 = no)		58.54
Use of barley in spring in the total flock diet (1 = yes, 0 = no)		31.71
Use of bran in spring in the total flock diet (1 = yes, 0 = no)		29.27
Use of agro-industrial by-products in spring in the total flock diet (1 = yes, 0 = no)		7.32
Use of natural rangeland in spring (1 = yes, 0 = no)		82.93
Use of barley in summer in the total flock diet (1 = yes, 0 = no)		100.00
Use of bran in summer in the total flock diet (1 = yes, 0 = no)		63.41
Use of straw in summer in the total flock diet (1 = yes, 0 = no)		41.46
Use of agro-industrial by-product in summer in the total flock diet (1 = yes, 0 = no)		12.20
Use of natural rangeland in summer (1 = yes, 0 = no)		46.34
		Mean of breeders who indicated yes
Cost of concentrate feed uses per year (USD)	Continuous	7321.05
Cost of bran feed uses per year (USD)		1568.19
Cost of straw uses per year (USD)		758.70
Cost of agro-industrial by-product uses per year (USD)		15.53
Cost of purchased water (USD)		254.67

$$Y_i = \alpha + \beta_1 x_{1,i} + \beta_2 x_{2,i} + \ldots + \beta_n x_{n,i} + \varepsilon, \quad (1)$$

where Y_i is the dependent variable that describes the supplemental feeding practices of breeder i; α is the regression constant; x is the explanatory variable; β is the coefficient of the explanatory variable; and ε is the error term.

Before running the logistic model, the problem of multicollinearity between the explanatory variables was tested, which makes it difficult to determine the effect of the explanatory variables on the dependent variable. This logistic regression is based on a maximum likelihood analysis, which is used to test the effect of the explanatory variables on the probability of the supplemental feeding practices of breeder.

As for the choice of explanatory variables in the model, including a systematic search was carried out for all variables statistically linked to the dependent variable. In our case, these variables were the attributes of the livestock feeding system, which were called "determinants" and likely to influence supplemental feeding practices. To detect the problem of multicollinearity between variables, statistical correlation tests should be followed. Thus, every time two correlated variables were found and one was deleted in order to solve this problem (Ahmed et al., 2018). In the end, the model included eight explanatory variables, which represented the factors that were assumed to influence the decision to supplemental feeding practices made by the breeders in the study area (Table 2).

3. Results

3.1. Livestock farming features of the El Ouara rangelands

3.1.1. Structure and management of livestock farming

The survey involved a sample of 50 breeders, 41.00% of whom practiced livestock farming and another economic activity second activity in parallel with. Breeders' average age was 64.82 years old. Household size was large with around six persons (Table 3). Herds were diversified, where 72.00% of the sample included mixed breeding of small ruminants (sheep and goats) and 36.00% of the respondents also raised camel livestock. Sheep species dominated herds, with only three breeders did not have small ruminants, these farms contained principaly camel livestock with an average of 63 camels. Herd size often varied between 15 and 1000 heads; 24.00% of breeders had a herd size of between 15 and 100 heads; 44.00% kept herds with 100–300 heads; and 32.00% of the respondents holded more than 300 heads, while 12.00% breeders had the herd size more than 500 heads. The main months of breeders spent on rangelands were September and February.

3.1.2. Herd guarding modality

The herd can be guarded in several ways by either its owner or another person. In the latter case, the owners can entrust their livestock to a salaried shepherd or to another breeder by means of a practicing association. Two owner categories with specific practices and strategies, "breeder" and "breeder-shepherd", were differentiated with 80.40% and 19.60% of the total sample, respectively.

The results revealed that only the breeders with a large number of livestock employed a permanent salaried shepherd. These breeders generally owned livestock more than 200 heads. During the calving period (about two to five months), the herd was often divided into two lots. The ewes in foal were left alone, while the rest of the flock formed another lot, when owners are obliged to employ two shepherds. Similarly, large herds that exceed 100 heads were most often kept by two people (a shepherd and a shepherd's assistant) throughout the year.

Shepherds were paid a monthly salary ranging from 231.00 to 330.00 USD, depending on the number of kept livestock, in addition to living costs, which varied between 66.00 and 99.00 USD per month. The shepherd also benefited on average from two months leave per year, which were not paid. In this case, the replacement shepherd was paid the salary. If shepherds had livestock, they did not pay the cost of herding but, in return, they paied the cost of supplementary feeding for their herds. All guarding contracts are ethical, and disputes will be settled by elderly, who are experienced people known for their impartiality.

It had two types of association practiced for breeders: "Cherka" and cash payment associations. The "Cherka" association involved another breeder joined by sharing herd guarding responsibilities, but they all had their own flock. This association type was used by

Table 2
Descriptions of the explanatory variables considered in the logistic model.

Explanatory variable	Description of explanatory variable	Type of explanatory variable	Expected sign
Household size	Household size of persons	Continuous	+ or –
Breeder age	Breeder age (years old)		–
Cereal area	Farm area used for cereal crops (hm^2)		+
Rangeland period	The period spent by the herd on rangelands (month)		+ or –
Herd size	Total herd size in livestock units (such as sheep, goat, and camels). Livestock unit (LU) coefficients for camels are 1.00 and for sheep and goats are 0.15		+ or –
Another economic activity	Breeder practicing another economic activity other than livestock farming activity (yes or no)	Categorical	+
Migrant remittance	Possession of income from emigrated family members (yes or no)		+
Agricultural development group membership	Breeder is an agricultural development group membership (yes or no)		+

Note: + or – means the variable can have a negative or positive impact on the supplemental feeding, respectively.

Table 3
General description of livestock farming features in the El Ouara rangelands.

Feature	Mean	Standard deviation
Average breeder age (years old)	64.82	15.50
Household size (persons)	5.74	2.94
Breeder practicing another economic activity (persons)	41.21	10.36
Permanent labour (persons)	1.80	0.40
Occasional labour (persons)	1.72	0.45
Camel size (LU)	23.50	42.05
Herd size (LU)	71.64	74.89
Rangeland period (month)	9.12	3.42
Transhumance period (month)	2.38	1.82
Ownership vans	1.12	0.33

10.00% of breeders. With the cash payment association, the partner payed money to the breeders every month according to the number of entrusted livestock. In general, the average price was 1.65–2.31 USD per head per month, and feed costs were added if necessary. It was the practice most widely used by breeders (53.00%). The remaining breeders did not use any association.

3.1.3. Rangeland use forms

In this unstable context characterised by marked climate and socio-economic changes, breeders undertook many practices according to their available means. Indeed, by identifying the different herd movement methods, we understood the main cattle management modes. The study showed three rangeland use forms, as shown in Table 4. Transhumant form means that herders move a long way depending on fodder and water availability, but without crossing Tunisian frontiers. A new transhumance mode over short distances has appeared, representing 58.00% of the sample. These breeders had the largest herd size. Sedentary form indicates that breeders live in the same land area all year round. Table 4 shows that only 34.00% of the sample undertook this practice in the same rangeland area all year round. Thirdly, semi-urban form denotes that some breeders living in town have developed a new management mode. They lead their herds to rangelands only during rainy periods. During long drought periods, 8.00% of breeders take herds home to urban areas. These individuals had the smallest herd size, which was limited to nine small ruminants on average with no camels.

3.2. Livestock feeding system

3.2.1. Importance of supplemental feeding on rangelands

The descriptive results of the total samples showed that breeders used a series of fodder resources to feed their livestock. Nonetheless, only 18.00% of breeders used the fodder resources of their livestock on natural rangelands all year round, and the other breeders provided their livestockwith supplemental feeding. Supplemental feeding refers to any feed products used by breeders to their livestock, rather than fodder naturally found by herds on rangelands. Barley was usually provided for flocks and is the first type of feed used by breeders when contemplating supplemental feeding. Bran is commonly incorporated with barley into the year-round feed, representing 10.00% of the total flock diet on average. The highest straw share was distributed in summer. Nevertheless, the proportions of the other agro-industrial by-products, principally olive cake, did not exceed 2.00% of the total flock diet. The cost of raising 65 heads was approximately 13,904 USD per year.

3.2.2. Types of livestock feeding system

The breeders who stated that they are totally based on grazing were not included in the supplemental feeding. Their feeding way constituted the first type of feeding system: natural rangeland feeding system, with 18.00% of breeders belonging to this system. The livestock in this cluster grazed all year on natural rangelands. The PCA was run for 41 breeders, and the variables used in the livestock feeding system are reported in Table 1.

The PCA results indicated that the first five principal components accounted for 66.21% of the total variation (Table 5). The first principal component (PC1) accounted for 20.54% of the total variation and was associated with indicative variables of a high feed supplementation cost (concentrate, straw, and bran) (Fig. 2). The second principal component (PC2) explained 16.44% of the variation and showed high agro-industrial by-product use (olive cake). The third principal component (PC3) represented 14.23% of the total variation. It referred to rangeland areas and high animal watering costs. The fourth principal component (PC4) explained 8.24% of the variation and showed that concentrate was used even in winter by some breeders. The fifth principal component (PC5) expressed the highest concentrate feed cost, which presented the importance of concentrate feed for flock diet and accounted for 6.76% of the total variation.

Table 4
Description of rangeland use forms.

Rangeland use form	Number of breeders	Percentage of the total samples (%)	Average number of small ruminants	Average number of camels
Transhumant form	29	58.00	51	45
Sedentary form	17	34.00	44	15
Semi-urban form	4	8.00	9	0

Table 5
Description of the first five principal components of livestock feeding system.

Principal component	Observed variable	Correlation coefficient	Percentage of explained variance (%)	Percentage of accumulated variance (%)
PC1	Cost of straw uses per year	0.49	4.31	20.54
	Cost of agro-industrial by-product uses per year	0.54		
	Use of barley in autumn in the total flock diet	0.69		
	Use of natural rangelands in autumn	0.53		
	Use of natural rangelands in winter	0.48		
	Use of barley in summer in the total flock diet	0.41		
	Use of agro-industrial by-product in spring in the total flock diet	1.56		
	Use of natural rangelands in summer	0.63		
PC2	Use of barley in spring in the total flock diet	1.05	3.45	36.98
	Use of agro-industrial by-products in spring in the total flock diet	2.22		
PC3	Cost of purchased water	0.60	2.99	51.21
	Use of bran in spring in the total flock diet	0.62		
PC4	Use of barley in winter in the total flock diet	1.96	1.73	59.45
	Use of natural rangelands in spring	0.75		
	Use of bran in winter in the total flock diet	1.88		
PC5	Cost of concentrate feed uses per year	0.47	1.42	66.21
	Use of barley in autumn in the total flock diet	0.32		
	Use of straw in autumn in the total flock diet	−0.28		

Note: PC1 means indicative variables of a high feed supplementation cost; PC2 shows high agro-industrial by-product use (olive cake); PC3 refers to rangeland areas and high animal watering costs; PC4 shows that concentrate was used even in winter by some breeders; PC5 expresses the highest concentrate feed cost.

We distinguished the remaining three livestock feeding systems based on the generated component. The secondirstwas the rangeland-based feeding system, representing only 10.00% of farms. The proportion of rangeland biomass in the flock diet in this cluster was the highest, at around 60.00% of the total flock diet. Livestock graze all year round on natural rangelands. In spring, feeding animals were based exclusively on rangelands. In summer, breeders supplemented their livestock with concentrate feeds (barley and bran). The third was the mixed feeding system, accounted for 36.00% of farms. Flocks feed on concentrate feeds and graze on rangelands (Table 6). The fouth was the concentrate-based feeding system, concerned 36.00% of farms. This group was characterised by using not only more supplemental feeds, which are based on concentrate feeds, but also agro-industrial by-product feeds (olive cake).

3.2.3. Factors influencing supplemental feeding on rangelands

The Wald chi-square tested the null hypothesis that the model coefficients are zero. The null hypothesis was rejected because the Wald chi-square was 62.63 (probability>Wald chi-square = 0.002), which indicated that at least one of the independent variables had a

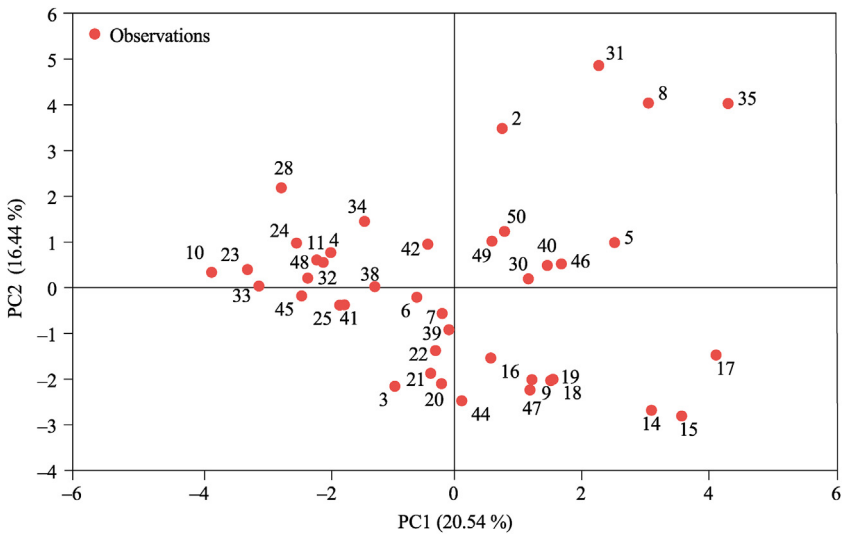

Fig. 2. Distribution of 41 breeders according to two principal components: PC1 and PC2. PC1, indicative variables of a high feed supplementation cost; PC2, high agro-industrial by-product use (olive cake).

Table 6
Statistics of the features of the four livestock feeding systems identified in the El Ouara rangelands.

Feature	Natural rangeland feeding system ($n = 9$)	Rangeland-based feeding system ($n = 5$)	Mixed feeding system ($n = 18$)	Concentrate-based feeding system ($n = 18$)
Camel size (LU)	26.33 ± 19.16	16.20 ± 15.58	24.83 ± 15.88	9.06 ± 13.26
Herd size (LU)	61.27 ± 45.85	80.46 ± 48.56	89.72 ± 83.18	46.66 ± 42.65
Rangeland period (month)	12.00 ± 0.00	9.20 ± 2.77	8.44 ± 4.00	6.22 ± 3.41
Transhumance period (month)	2.22 ± 1.48	3.40 ± 2.41	1.02 ± 0.67	0.93 ± 0.52
Ownership vans	0.89 ± 0.33	1.00 ± 0.00	0.89 ± 0.32	0.83 ± 0.51
Family labour (persons)	1.40 ± 0.55	1.22 ± 0.44	1.39 ± 0.61	1.33 ± 0.59
Permanent labour (persons)	0.22 ± 0.44	0.40 ± 0.55	0.17 ± 0.51	0.22 ± 0.43
Occasional labour (persons)	0.33 ± 0.50	0.60 ± 0.55	0.28 ± 0.57	0.22 ± 0.43
Watering cost (USD)	238.22 ± 265.38	85.86 ± 168.59	96.00 ± 214.66	467.55 ± 1027.70
Concentrate cost (USD)	0.00 ± 0.00	5447.46 ± 6751.65	7559.80 ± 6831.53	13,206.40 ± 8791.04
Straw cost (USD)	0.00 ± 0.00	370.98 ± 784.27	529.00 ± 881.28	2754.56 ± 3450.66

Note: Mean ± SD, mean values and standard deviation.

significant influence on the dependent variable (supplemental feeding). Further, a good model fit was demonstrated using Pseudo R^2 McFadden. In fact, Pseudo R^2 McFadden had a satisfied high value, equaling 0.83, which proved the model's performance. The logistic regression results showed that six variables including breeder age, cereal area, herd size, migrant remittance (significant at the 5.00% level), another economic activity, and agricultural development group membership (significant at the 10.00% level) had significant impact on supplemental feeding (Table 7).

4. Discussion

This study was conducted on the common rangelands located in El Ouara of the Ben guarden District of southern Tunisia, characterised by livestock farming, which has always been considered as a risky activity (Faye, 2001). To ensure the sustainability of the livestock feeding system when facing climate change, supplemental feeding on rangelands is one of the strategies followed by breeders.

4.1. Diversity of grazing practices and rangeland uses

Today, the cash payment association type has become an ordinary practice used by breeders (53.00%). Thus, it is cheaper for breeders to give their herd to an association than to recruit a shepherd, with a herd size varying between 15 and 100 heads, accounting for 24.00% of the respondents. In parallel, other studies have indicated that herd size on rangelands has reduced from an average of 500 heads to an average of 300 heads in flocks (e.g., Elloumi et al., 2001).

In the past, we organised transhumance according to fodder availability without considering regional boundaries or country borders. Our study proved that even today, transhumance is still an important grazing practice in the El Ouara rangelands (58.00% of the respondents used transhumant form), but it only covers short distances without crossing Tunisian borders. As reported by Bernués et al.

Table 7
Logistic model estimation results of the selected eight variables.

Variable	β	Standard deviation	Wald	*P*-value	Odds ratio
Constant	14.17	9.24	2.35	0.13	–
Household size	−0.09	0.57	0.02	0.88	0.92
Breeder age	−0.20	0.10	3.89	0.05**	0.82
Cereal area	0.98	0.51	3.63	0.06**	2.65
Rangeland period	−0.40	0.35	1.31	0.25	0.67
Herd size	−0.09	0.05	4.14	0.04**	0.91
Breeder practicing another economic activity (yes)	1.57	0.94	2.77	0.10*	4.79
Breeder practicing another economic activity (no)	−1.57	0.94	2.76	0.10*	0.21
Migrant remittance (yes)	−3.90	1.81	4.64	0.03**	0.02
Migrant remittance (no)	3.90	1.81	4.64	0.03**	49.60
Agricultural development group membership (yes)	2.49	1.44	2.99	0.08*	12.06
Agricultural development group membership (no)	−2.49	1.44	2.99	0.08*	0.08
Wald chi-square	58.17				
Pseudo R^2 McFadden	0.83				
−2 Log (Likelihood)	39.53				

Note: *, significant at the 10.00% level; **, significant at the 5.00% level; β, regression coefficient, indicating how much the value of a dependent variable varies with a change in the value of the independent variable; -, no data.

(2011), the livestock feeding system traditionally based on rangelands in southern Tunisia has undergone a series of changes, such as the disappearance of transhumance whether towards neighboring countries (Libya and Algeria), or towards the central and northern parts of the country. The sedentary form was also detected as a grazing practice. Breeders stated that because of the harsh climate condition, it is better to stay in the same place all year round generally near a well. In parallel to these traditional grazing practices, the development of the semi-urban form has taken place, which is closely linked with the city (Gaddour et al., 2014).

4.2. Supplemental feeding: a pillar of the livestock feeding system on rangelands

As rangeland vegetation in the region is typically steppe, fodder production varies according to precipitation. As a result, breeders follow supplemental feeding on rangelands as a strategy to face scarce natural fodder resources. This practice appeared with drought in the 1970s and was performed by the Tunisia to face the strong aridity (Abaab and Genin, 2004). Since then, to compensate for lack of fodder and (or) nutritional deficit, breeders have resorted to supplemental feeding. Links of the livestock feeding system with fodder concentrates, agro-industrial by-products, and crop waste have increased (Ibidhi and Salem, 2018). Since Tunisia is an importer of cereals, the livestock feeding system has been linked to the international cereals market.

Generally, supplemental feeding is very seasonal, occurring mostly during the dry season due to the scarcity of natural rangelands and a drastic drop in nutritional quality. Today, this practice is generalised among breeders almost all year round, as our results proved. Thus, only 18.00% of the respondents did not use supplemental feeding. Rrangeland-based, mixed, and concentrate-based feeding systems were based on concentrate use in summer. Moreover, the breeders using the concentrate-based feeding system implemented a type of livestock intensification on rangelands by keeping their herds on rangelands, while distributing feed all year. As a result, breeders practiced many conduct modes, which varied from conserving extensive livestock feeding system on rangelands to semi-extensive up to intensive ones. Gaddour et al. (2014) proved that livestock feeding system is generally declining to agropastoralism, and expanding agriculture has been observed in southernTunisia (Jaouad et al., 2022).

Our results showed that the breeders with the largest camel size still used natural rangeland feeding system, while the breeders with the largest herd size adopted a mixed feeding system (Table 6). Consequently, the supply of barley and coarse feed (hay and straw) has become a habit, and usages annually increased. Similar results have been found in other studies (Nasr et al., 2000; Guillaume, 2009) performed on Tunisian rangelands, with only a minority of large breeders still using natural rangeland feeding system in the El Ouara rangelands, eastern Jeffara, or Dahar. Feed intensification is also a common trend in agricultural and livestock feeding systems in most regions of the world (Powell et al., 2004; Vall et al., 2017; Godde et al., 2019; Carvalho et al., 2022).

This study revealed that supplemental feeding was determined by several factors. Thus, a set of structural and socio-economic determinants may influence supplemental feeding in the El Ouara rangelands. A positive correlation was detected between the cereal area and supplemental feeding. Consequently, the probability of adopting this practice increased with the increase of cereal area, which was linked with herd size. The breeders with a limited number of herds were more likely to supplement their herd because of high livestock feed cost. In some cases, breeders resorted to supplementing only new borns and/or herds with poor health. Practicing a secondary livestock feeding system also had a significant positive effect on supplemental feeding. In parallel, this study showed that the migrant remittance had a significant positive effect on supplemental feeding. Therefore, acquiring a funding source is very important for breeders to feed livestock and increase production, while being an agricultural development group membership can promote supplemental feeding. Thus, belonging to an agricultural development group membership is important for ensuring the feed needed for flocks. In addition, young breeders are more inclined to supplemental feeding than older breeders. In line with this finding, several studies have shown the weight of socio-economic factors on integrating new farming practices (Kanoun, 2016; Montcho et al., 2018).

4.3. Degradation of the El Ouara rangelands

Nowadays, the intensification of livestock feeding system caused by supplemental feeding has completely changed the behavior of breeders on rangelands, with consequences including the degradation of fodder potential. In addition, the El Ouara rangelands are subject to different management methods, and the exploitation of most rangelands is common. Therefore, breeders have free access to the pastoral resources in these areas. Due to the lack of restrictions to the number of herds, overloading of livestock on rangelands has led to the degradation of natural fodder resources. Moreover, breeders compete for the use of natural pasture fodder because every one has the right to use it. Consequently, the pastoral ecosystem has continuously deteriorated for several decades (Jeder and Sghaier, 2010).

5. Conclusions and implications

Breeders attempt to adapt their practices to climate change to ensure the sustainability of pastoral ecosystem. This study revealed the development of semi-urban form closely linked with the city. The multivariate statistical analyses identified four different feeding systems: natural rangeland, rangeland-based, mixed, and concentrate-based feeding systems. The connections among different rangeland use forms and feeding systems represent an essential step to develop strategies for a more sustainable livestock feeding system in Tunisia. Supplemental feeding is increasingly adopted by breeders to overcome the fodder deficit in the El Ouara rangelands. This practice is determined by a combination of several factors. Logistic regression shows that six significant variables (breeder age, cereal area, herd size, migrant remittance, second activity, and agricultural development group membership) affect supplemental feeding. The financial means is decisive for adopting supplemental feeding, such as practicing a second activity or possessing migrant remittance. Young breeders are more inclined to practice supplemental feeding than older breeders. Barley and bran are the main supplement feed

for livestock. As a result, the livestock feeding system is becoming increasingly independent. Moreover, the intensification of the livestock feeding system leads to the degradation of the fodder potential of rangelands. However, to ensure the sustainability of the livestock feeding system and meet the growing demand of livestock farming products, approaches such as new farming practices and new feeding technologies are recommended to be adopted.

Authorship contribution statement

Houda Rjili: conceptualisation, investigation, data curation, formal analysis, methodology, software, and writing- original draft. Mohamed Jaouad: resources, supervision, writing-review and editing, and validation. Chaker Selmi: data presentation, software, and methodology.

Ethics statement

Ethics approval was obtained from the Ethics Committee of Arid Regions Institute of Medenine. In addition, the participants provided their informed consent to participate in this study.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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